# Using Natural Resources for Energy

*Unit 2 Study & Review · Using Natural Resources for Energy · Teacher Key*

**Name:** _____ **Date:** _____

## A · MATCH THE WORD

1-H · 2-D · 3-G · 4-C · 5-F · 6-E · 7-A · 8-B

## B · RENEWABLE OR NONRENEWABLE?

coal N · wind R · petroleum N · natural gas N · sunlight R · uranium N · flowing water R · geothermal heat R

## C · FILL THE PATHWAY

burn the fuel → heat water into steam → spin a turbine → turn a generator → electricity
Wind: moving air → blades (turbine) → generator → electricity. Solar: sunlight → solar cell → electricity.

## D · COMPARE

renewable: as quickly as it is used. nonrenewable: as quickly as it is used (cannot be restored quickly). abundant but still: nonrenewable.
Three fossil fuels: coal, petroleum, and natural gas. Nuclear fuel: uranium.

## E · LABEL IT

fuel (coal or nuclear) → steam (or water) → turbine → generator → electricity. (Boxes: steam/water, turbine, generator, electricity.)

## F · SCENARIO

1. Wind. 2. Strong, steady wind most of the year. 3. Wind can be calm some days / needs a windy location. (Accept any reasoned choice with one fact + one limitation.)

## G · SHORT RESPONSE

Any resource: one benefit + one limitation + a location/condition where it works well.
Examples: wind — benefit: no fuel to burn; limitation: needs wind; works well where it is windy. Solar — benefit: sunlight is renewable; limitation: cloudy days/shade; works well in sunny places.